Lecture 16

AUTO REGRESSION (AR)

ARIMA

AR — MOVING AVERAGE

I — INTEGRATED

Data: y_t $(y_1 \dots y_n)$

AR(p) → order

$$y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

$\varepsilon_t \sim N(0, \sigma^2)$

$t = p+1, p+2, \dots, n$

(Regression with $n-p$ observations)

$$y = X\beta + \varepsilon$$

$$y = \begin{pmatrix} y_{p+1} \\ \vdots \\ y_n \end{pmatrix}, \quad X = \begin{bmatrix} 1 & y_p & y_{p-1} & \cdots & y_1 \\ \vdots & y_{p+1} & y_p & & \vdots \\ \vdots & \vdots & \vdots & & \\ 1 & y_{n-1} & y_{n-2} & & y_{n-p} \end{bmatrix}$$

$$\beta = \begin{pmatrix} \phi_0 \\ \vdots \\ \phi_p \end{pmatrix} \qquad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

sm.ols(y, x) → $\hat{\beta} = (X^T X)^{-1} X^T y$

$\hat{\phi}_0, \hat{\phi}_1 \cdots \hat{\phi}_p$

$$y_{n+1} = \hat{\phi}_0 + \hat{\phi}_1 y_n + \hat{\phi}_2 y_{n-1} + \cdots + \hat{\phi}_p y_{n+1-p} + \varepsilon_{n+1}$$

($\varepsilon_{n+1} \to 0$)

$$\hat{y}_{n+1} = \hat{\phi}_0 + \hat{\phi}_1 y_n + \hat{\phi}_2 y_{n-1} + \cdots + \hat{\phi}_p y_{n+1-p}$$

$$\hat{y}_{n+2} = \hat{\phi}_0 + \hat{\phi}_1 \hat{y}_{n+1} + \hat{\phi}_2 y_n + \cdots + \hat{\phi}_p y_{n+2-p}$$

$\hat{y}_{n+3}, \hat{y}_{n+4}$ ... recursive prediction.

$$\hat{y}_{n+j} = \hat{\phi}_0 + \hat{\phi}_1 \hat{y}_{n+j-1} + \cdots + \hat{\phi}_p \hat{y}_{n+j-p}$$

where $\hat{y}_i = y_i$ for $i = 1, \ldots, n$

History of AR

YULE (1927) invented AR for the sunspots data.

y_t : annual sunspots

$$y_t = \beta_0 + \beta_1 \cos 2\pi f t + \beta_2 \sin 2\pi f t + \varepsilon_t$$

→ Lecture 7

$\beta_0, \beta_1, \beta_2$ & f

f → corresponded to a period of 11 years.

interval estimate narrow around 11 years

First Observation

data = sinusoid + noise

$$s_t = \beta_0 + \beta_1 \cos 2\pi f t + \beta_2 \sin 2\pi f t$$

$$y_t = s_t + \varepsilon_t$$

(s_t: sinusoid, ε_t: noise)

$$\boxed{s_t = \beta_0 + \beta_1 \cos 2\pi f t + \beta_2 \sin 2\pi f t}$$

$$s_t'' = \frac{d^2}{dt^2} s_t = -(2\pi f)^2 \left[\beta_1 \cos 2\pi f t + \beta_2 \sin 2\pi f t\right]$$

or,

$$\boxed{s_t'' = -(2\pi f)^2 (s_t - \beta_0)}$$

For us, this becomes (discrete time)

$$\boxed{(s_t - s_{t-1}) - (s_{t-1} - s_{t-2}) = 2\left[\cos(2\pi f) - 1\right](s_{t-1} - \beta_0)}$$

$s_t = \beta_0 + \beta_1 \cos 2\pi f t + \beta_2 \sin 2\pi f t$ satisfies

Moreover, every s_t satisfying the difference equation is a sinusoid for some β_1 & β_2.

$$s_t = \beta_0 + \beta_1 \cos 2\pi f t + \beta_2 \sin 2\pi f t$$

$$\Updownarrow$$

$$s_t - 2s_{t-1} + s_{t-2} = 2(\cos 2\pi f - 1)(s_{t-1} - \beta_0)$$

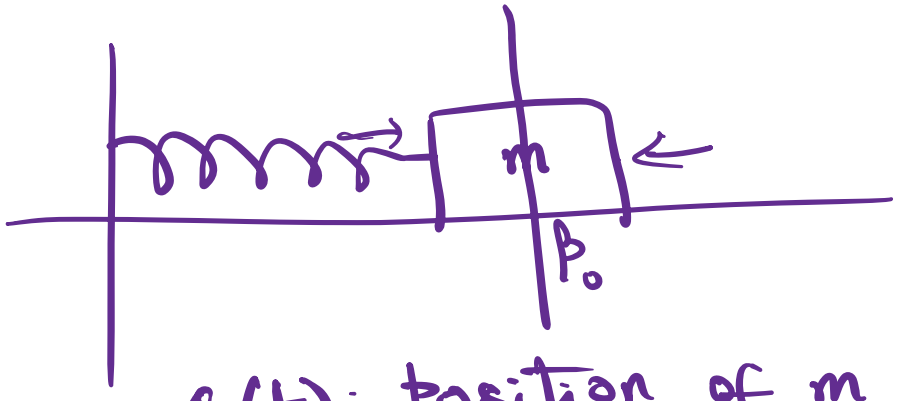
$$y_t = \underbrace{\beta_0 + \beta_1 \cos 2\pi f t + \beta_2 \sin 2\pi f t}_{s_t} + \varepsilon_t$$

$$y_t - 2y_{t-1} + y_{t-2} = 2(\cos 2\pi f - 1)(y_{t-1} - \beta_0) + \varepsilon_t$$

$$y_t = \phi_0 + \phi_1 (y_{t-1} - y_{t-2}) + \varepsilon_t$$

$$y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t \rightarrow AR(2)$$

special case of AR(2) with $\phi_2 = -1$



$s(t)$: position of m at time t

Force on m at time t

$$= -K(s(t) - \beta_0)$$

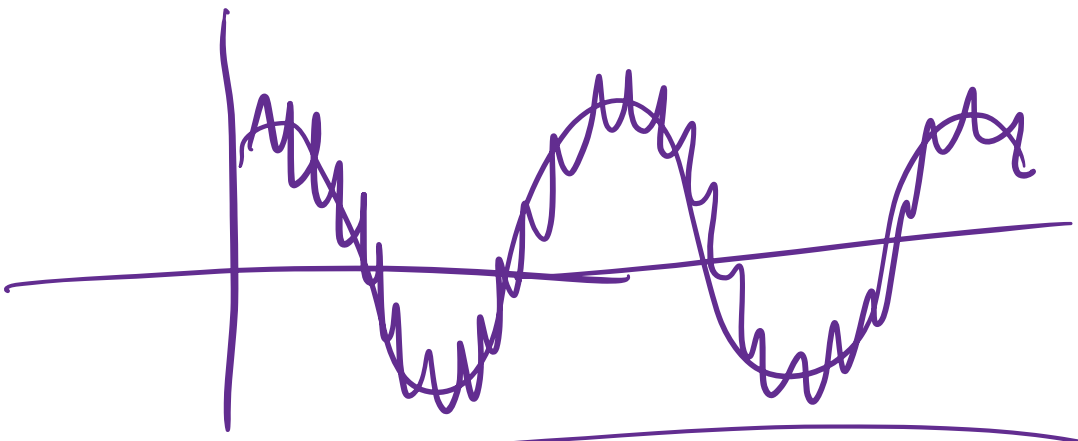
$$-K(s(t) - \beta_0) = m s''(t)$$

$$s''(t) = -\frac{K}{m}(s(t) - \beta_0)$$

$$s''(t) = -(2\pi f)^2 (s(t) - \beta_0)$$

$$\frac{K}{m} = (2\pi f)^2$$

Model 1: $y_t = s_t + \varepsilon_t$ → Noisy measurements



Model 2:

$$s''(t) = -\frac{K}{m}(s(t) - \beta_0) + \text{noise}$$

→ motion itself is perturbed.

Yule Model:

$$\longrightarrow y_t = \phi_0 + \phi_1 y_{t-1} - y_{t-2} + \varepsilon_t$$

$$\text{AR(2)}: \quad y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t$$

Regression of $y_t + y_{t-2}$ on y_{t-1}

$$\phi_1 = 2\cos(2\pi f)$$

Linear Regression Example

$$y_t = \log(CPI_t)$$

$$y_t = \underbrace{\beta_0 + \beta_1 t}_{m(t)} + \varepsilon_t$$

$100 \times \beta_1$: Percent inflation (monthly)

$12 \times 100 \times \beta_1$: annual inflation

$\longrightarrow$ [3.749%, 3.84%]

$$y_t = \underbrace{m(t)}_{\text{linear}} + \varepsilon_t$$

$$m(t) = \beta_0 + \beta_1 t$$

$$\updownarrow$$

$$m'(t) = \beta_1$$

$$m_t - m_{t-1} = \beta_1$$

$$y_t - y_{t-1} = \beta_1 + \varepsilon_t$$

$\hat{\beta}_1$ = ~~Average~~ of $(y_t - y_{t-1})$